

NETFLIX PRIZE DATASET

Personalized Movie Recommendations

Three collaborative-filtering models. Two very different report cards.
We measured rating accuracy (RMSE) and ranking quality (MAP@10) — and discovered that being good at one says surprisingly little about the other.

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THE DATA AT A GLANCE

24.0M

movie ratings
loaded

470.7k

unique users

4,499

movies (file 1)

98.9%

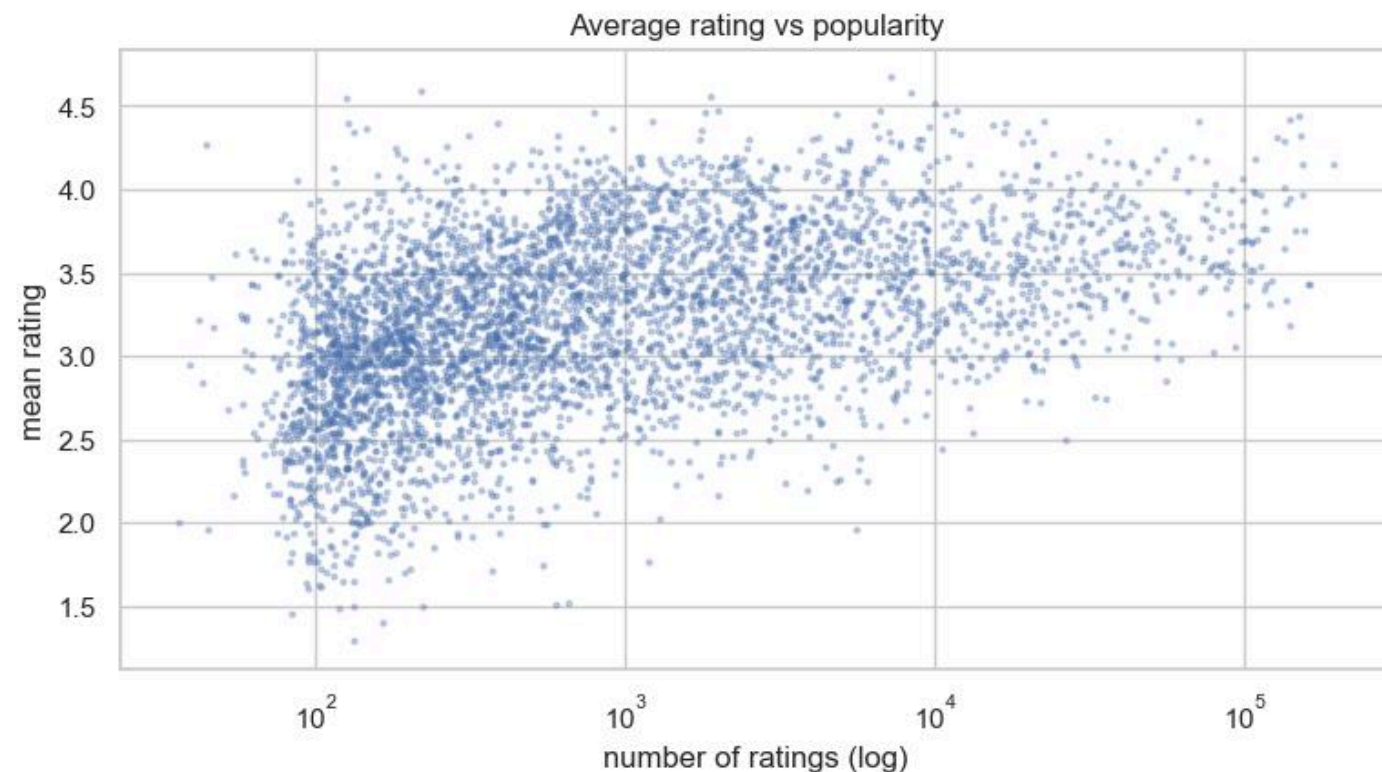
of the matrix is
empty

Sparsity is the whole game. Each user rated a tiny fraction of the catalog — the model must infer taste from almost nothing. That is exactly what collaborative filtering was invented for.

What the data told us

Two findings shaped every modeling decision we made downstream.

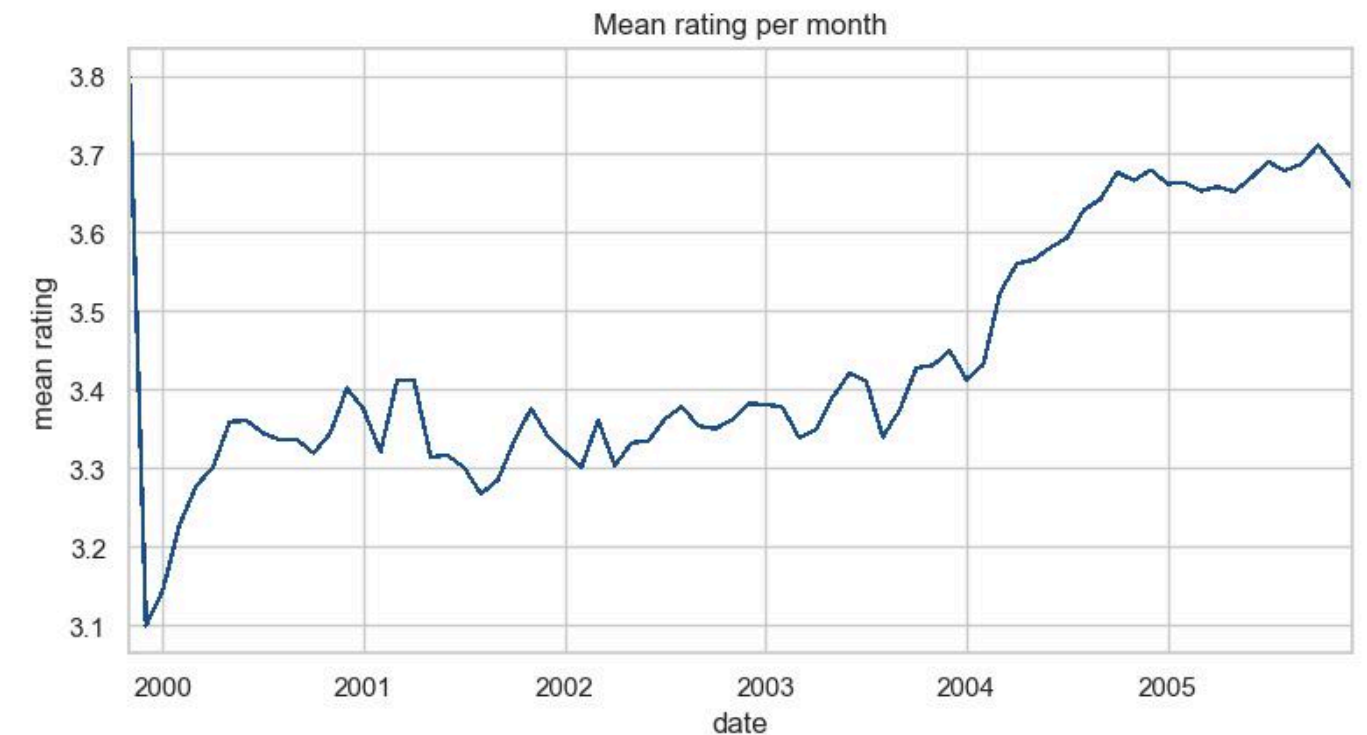
Tiny samples, extreme averages



A 4.8 from 40 fans is not better than a 4.2 from 100k people.

We flagged this funnel as a trap — and it later broke our first ranking evaluation.

Rating behaviour shifts around 2004



56.5%

of ratings already count as relevant (≥ 3.5) — so we rank the full catalog

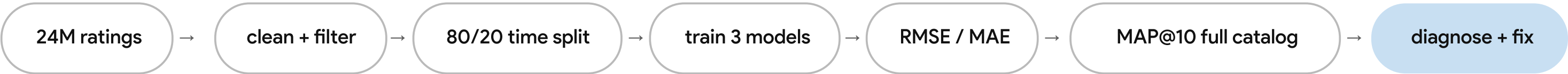
80 / 20

time-based split per user — random splits would leak the future

Our approach

Classical collaborative filtering — deliberately. Simple, defensible, and what the Netflix Prize itself proved out.

PIPELINE



Baseline

global mean + user bias + item bias
Zero personalization — it predicts the same relative ordering for everyone. It exists as a measuring stick: any smart model must beat it, or it learned nothing.

0.9250

RMSE · REFERENCE MODEL

Item-KNN

k = 40 · pearson-baseline · WithMeans
“People who liked X also liked Y.” Item-based, because the item×item matrix fits in memory; WithMeans, because users rate on different scales.

0.8959

RMSE · BEST EXPLAINABILITY

SVD

100 factors · 25 epochs · reg 0.05
The Netflix Prize approach. Learns a latent taste vector per user and per movie — a prediction is simply how well the two profiles match.

0.8885

RMSE · WINNER ON ACCURACY + SPEED

Rating prediction

Both trained models beat the baseline. SVD wins on accuracy, speed, and scalability.

Test set: 668,282 held-out future ratings

Model	RMSE ↓	MAE ↓	Fit	Predict
Baseline	0.9250	0.7256	3.8 s	—
Item-KNN	0.8959	0.6877	23.6 s	34.5 s
SVD	0.8885	0.6923	15.8 s	2.9 s

Why KNN wins MAE but loses RMSE. KNN is slightly better on typical predictions but makes

a few large mistakes on movies sharing few raters with the user's history — and RMSE punishes large errors hard. SVD's regularization keeps every error controlled.

0.8885

best RMSE — SVD matrix factorization

12×

faster prediction than KNN (2.9 s vs 34.5 s) and it scales linearly

−4.0%

RMSE vs baseline. Small number, real impact — Netflix paid \$1M for 10%

THE SURPRISE

Our best model lost to a popularity list.

Full-catalog Top-10 ranking · 1,000 users · relevance = test rating $\geq$ 3.5. **Accurate ratings $\neq$ good recommendations.**

0.0439

MAP@10 — popularity ranker, zero ML

0.0094

MAP@10 — SVD raw · 4.7× behind

0.0008

MAP@10 — Item-KNN raw · 55× behind

Diagnosis — the niche-movie trap.

10.1% of all recommendation slots (201 of 2,000) went to movies with under 100 ratings but extreme averages — the same few titles for everyone: Inu-Yasha: The Movie 3 (24 ratings), Combat! S2 (10), My Fair Lady: Bonus Material (15). The model is over-confident about items it has barely seen.

Filter, then personalize

Only movies with $\geq \text{min_pop}$ training ratings enter the candidate pool. The model personalizes within it.

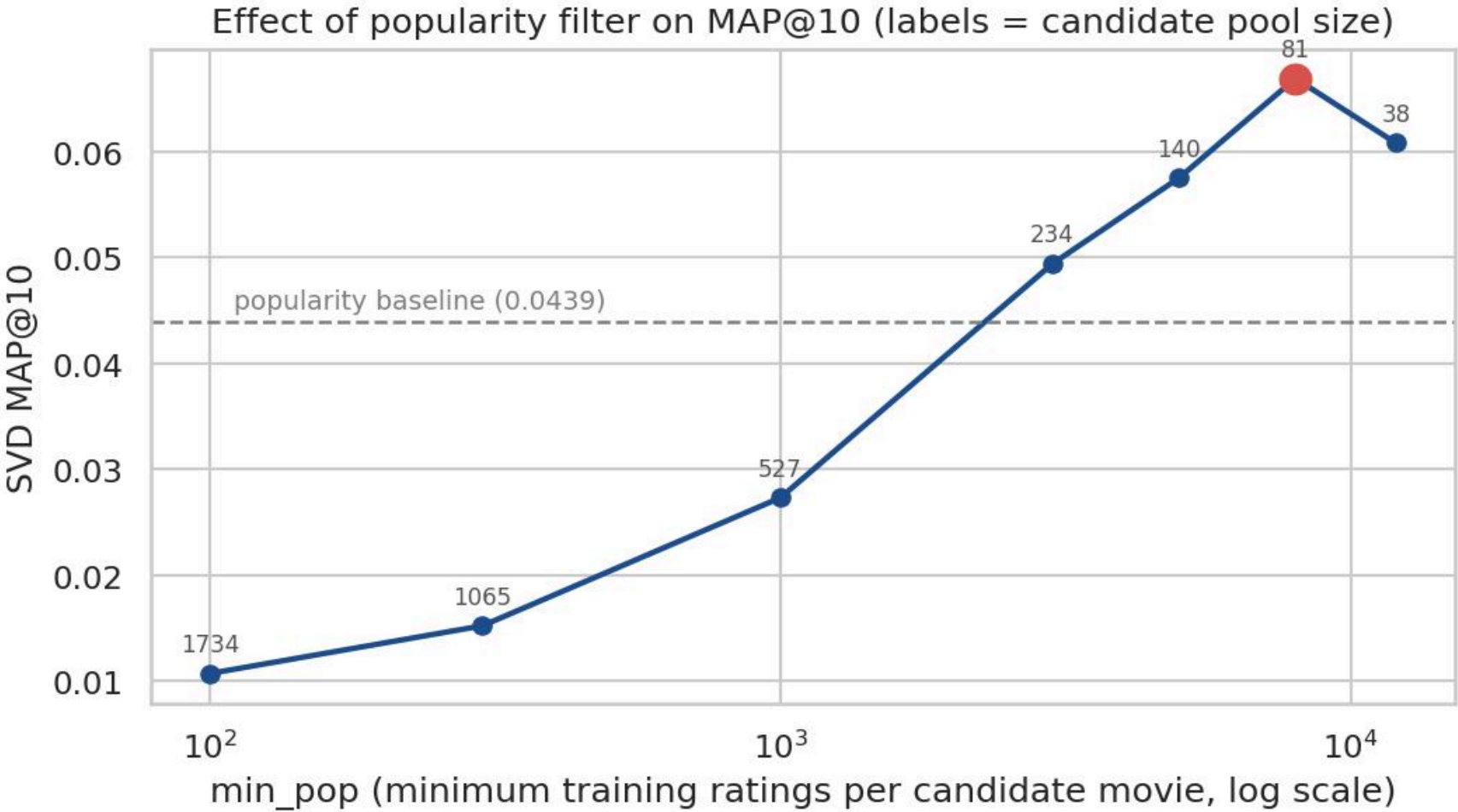
0.0669

SVD MAP@10 after filtering — was 0.0094 raw

+52%

vs the popularity baseline (0.0439). Filtered KNN also clears it (0.0643)

Honest note. min_pop was tuned on the same 1,000 evaluation users — a separate validation set would be more rigorous. We state it rather than hide it.



Sweep result: MAP@10 climbs as rare movies are removed, peaks at min_pop = 8,000 (81 candidates), then collapses once the pool is too small to personalize.

What it actually recommends

Metrics hide things — so we read the real Top-10 lists.

Active user • 3,016 ratings • SVD Top-5

Recommendation	Pred	Pop	Hit
Inu-Yasha	5.00	181	
Lost: Season 1	5.00	177	
Inu-Yasha: The Movie 3	5.00	24	✓
Mobile Suit Gundam SEED	4.99	56	
Nip/Tuck: Season 2	4.95	251	

Success: real personalization. The anime hit had only 24 training ratings — a popularity system could never surface it. The model also learned user bias: for a harsh rater (mean 1.30) it predicts ~3.0, not naive 5s.

FAILURE CASES

Sequential content

Recommends Six Feet Under: Season 4 to someone who never saw. Seasons 1–3. CF sees rating patterns, not story order.

Bonus-material trap

GoodFellas: Bonus Material ranks high — tiny devoted audiences inflate averages. The niche trap in another costume.

No diversity control

One list held several Simpsons seasons at once. Items are scored independently — nothing penalizes repetition.

TAKEAWAYS

- 01 Good RMSE $\neq$ good rankings.**
Our most accurate model initially lost to a trivial popularity list on Top-10 quality.
- 02 Measure, then fix.**
Quantifying the niche trap (10.1% of slots) led to a filter that beat the baseline by 52%.
- 03 Every fix has a cost.**
The same filter killed our best niche discovery and cut coverage from 6.2% to 1.7%.
- 04 Look at real outputs.**
Sequential seasons, bonus discs, franchise spam — no offline metric caught any of it.

What's next

Hybrid + metadata

fixes sequential seasons, helps cold-start movies

Diversity re-ranking

one franchise can't fill the Top-10

Confidence-aware scores

keep niche gems without the niche trap

ALS at full 100M scale

built for exactly this size

Fully reproducible

Two documented notebooks, fixed seeds, parquet pipeline.

[\[GitHub repository link\]](#)